

A Brief Review on Quantum-Based Control Strategies for Robotic Systems

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Abstract— The increasing convergence of quantum computing (QC) and advanced robotic control is giving rise to a new class of intelligent automation systems. Utilizing quantum-inspired computational principles, contemporary robotic platforms demonstrate greater optimization efficiency, superior perceptual processing, and improved decision-making performance. This survey outlines the principal quantum-oriented control methodologies utilized in robotics, with particular emphasis on approaches that incorporate Quantum Neural Networks (QNN), Quantum Machine Learning (QML), and Quantum-behaved Particle Swarm Optimization (QPSO). The investigation systematically reviews recent advances in quantum-assisted adaptive control, trajectory planning, and optimization frameworks, spotlighting their contributions to improved robustness, precision, and energy efficiency. By analyzing research in quantum reinforcement learning, distributed Noisy Intermediate-Scale Quantum (NISQ) computing, and hybrid quantum-classical approaches, this review clarifies how quantum principles are transforming both theoretical and practical aspects of robotic control. Likewise, several real-world implementations ranging from industrial manipulators to autonomous and cognitive robotic systems demonstrate the growing feasibility of quantum-enhanced decision-making. The review concludes that while remarkable progress has been made in quantum-inspired control for robotics, most existing studies remain constrained to simulations with restricted experimental validation on physical platforms. Addressing this gap will be essential in future research to transition from theoretical and simulated models toward hardware-level implementations, enabling the realization of scalable, fault-tolerant quantum control architectures.

Keywords— *Quantum Computing, Quantum control strategies, Quantum AI, Quantum Algorithms, Robotic systems*

I. INTRODUCTION

The quest for quantum-inspired control mechanisms in robotics marks a transformative shift toward integrating quantum principles to enhance precision and efficiency. Quantum-inspired techniques are increasingly important in robotics because they tackle high-dimensional optimization problems more efficiently than classical methods. Whether

implemented as hybrid classical–quantum schemes or fully quantum models, these approaches leverage principles such as superposition and entanglement to accelerate information processing critical for tasks such as real-time decision-making, resource allocation, and robotic path planning.

Fusion of quantum physics with control theory enhances understanding and management of quantum dynamics through diverse models, extending quantum mechanics' benefits to medical imaging and surgical precision while complementing foundational theories [1]. Expanding QC applications in robotics, Korenkov et al. [2] addressed gaps in quantum software engineering, focusing on quantum algorithmic gate design and applications, thereby improving robotic control and supporting the healthcare advances in [1]. Tamizi et al. [3] noted a significant rise in studies on QC and Deep Learning (DL), emphasizing their role in accelerating tasks like pattern recognition vital to intelligent robotics. Industrial studies [4-6] demonstrated quantum-inspired strategies in motor control and energy optimization, achieving greater precision, reduced torque ripple, and improved efficiency in robotic and electromechanical systems. These techniques address classical inefficiencies in dynamic, high-dimensional environments, promoting hybrid models combining classical and learning-based strategies. In quantum networks and multiagent systems, Yun et al. [7] proposed Quantum Multiagent Actor-Critic Neural Networks to coordinate multi-robot systems, improving coordination and computation for scalable Industry 4.0 solutions. Similarly, Khoshnoud et al. [8] explored integrating quantum technologies, including quantum entanglement and cryptography, into unmanned autonomous systems to enhance security and operational efficiency in networked dynamics. Likewise, integrating quantum control methodologies derived from robotic trajectory optimization into quantum optimal control schemes demonstrates how cross-disciplinary innovations can leverage complex robotic dynamics to refine quantum control approaches [9].

This review comprehensively synthesizes literature on QC-based control in autonomous systems and robotics. It examines the impact of QML and neural networks, analyzes quantum algorithms for robotic control and decision-making, and explores practical implementations and future applications of

quantum control in robotics. The scope includes studies involving autonomous, mobile, humanoid robots, manipulator arms, and autonomous underwater vehicle (AUV).

The present analysis contributes to the field by comprehensively synthesizing existing research on integrating QC into robotic system control. It reviews the principal methodologies and frameworks that employ quantum algorithms in autonomous and robotic platforms, outlines prospective research avenues, and identifies emerging applications of quantum-driven and quantum-inspired control techniques within robotics. The document is structured as follows: Section I introduces the topic and context. Section II outlines the literature selection and analysis methodology. Section III explores QNN in robotics. Section IV discusses QML for autonomous system control. Section V reviews approaches and their effectiveness in quantum-based algorithms for robotics control. Section VI presents practical applications of quantum control. Section VII provides critical analysis. Finally, Section VIII summarizes results and proposes future research directions.

II. METHODOLOGY

The search query “quantum computing” AND “robotics” was applied to specific fields, covering the timeframe from January 2015 to October 2025. The bar chart in Fig.1 reveals that China (n=8,349) and the United States (n=5,488) are the foremost contributors, followed by India, the United Kingdom, and Germany. The figure reflects raw publication counts, not accounting for the overall output in robotics and AI, which should be considered when making comparisons among countries. While variations in database coverage and indexing practices may influence absolute values, the overarching trend clearly highlights the prominent role of China and the United States in the advancement of QC applications within robotics.

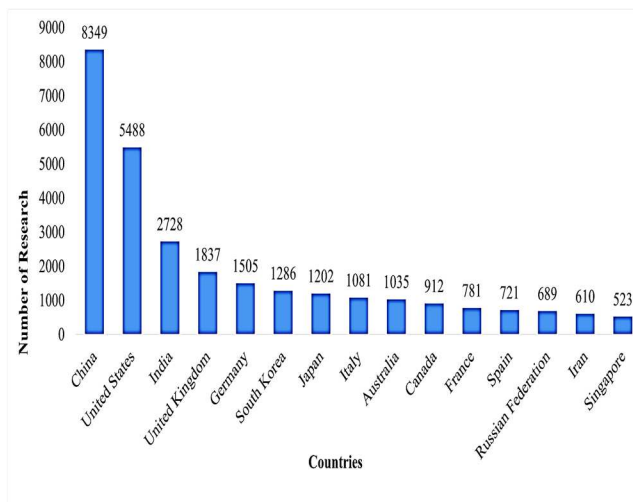


Fig. 1. Country-level contributions to quantum-computing-in-robotics publications; Data from Scopus and Web of Science (retrieved Oct. 17, 2025).

This review presents a structured analysis of the intersection between QC, artificial intelligence (AI), and robotics,

summarizing key advancements, challenges, and future directions. The search strategy was carefully designed to cover a wide range of relevant literature, including influential studies on integrating QC and AI in robotic systems. Comprehensive searches were conducted across major electronic databases, IEEE Xplore, Scopus, ScienceDirect, SpringerLink, Web of Science, PubMed, and the ACM Digital Library, recognized for their extensive robotics, computer science, and engineering collections

The search was limited to materials published between January 2015 and October 2025 to focus on recent and relevant advancements. Only English texts accessible in full through institutional subscriptions or open-access platforms were included. This period and language selection aimed to capture cutting-edge developments in QC applications in robotics while ensuring source relevance and manageability.

III. QNN-BASED CONTROL

The significant advancements in robotics through quantum computation envision more intelligent robots leveraging QNNs in AI and diagnostics [10-12]. Integrating quantum concepts with neural optimization deepens through neural-enhanced frameworks, which solve constrained optimization problems on quantum architectures and create new optimization opportunities for robotics [13].

Embedding quantum perceptrons and quantum neural networks (QNNs) within quantum computing (QC) provides a basis for exploring novel robotic applications and advancing quantum technologies [14]. Numbi, Fazilat, and Zioui introduced a quantum-inspired hybrid artificial neural network (QANN) to identify dynamic parameters of car-like mobile robots. Their study compared conventional artificial neural networks (ANNs) with QANNs in estimating key physical properties, such as mass and moment of inertia. Results demonstrated that QANN reduced training root mean square error (RMSE) by 9.7% and validation RMSE by 84.4% compared to the classical ANN model. In addition to improved accuracy, QANN showed greater robustness and superior generalization to previously unseen data, highlighting its potential for real-time modeling, trajectory prediction, and control of autonomous mobile robots [15].

Hanna et al. [16] proposed an adaptive PID controller based on a quantum neural network (APIDC-QNN) for the control of nonlinear systems. The application of the Lyapunov stability framework facilitated stable weight adaptation, efficient optimization, and rapid convergence. The APIDC-QNN achieved superior set-point tracking and demonstrated robust resistance to parameter variations, resulting in the lowest Integral Absolute Error (IAE) and Mean Absolute Error (MAE) in all evaluations. These findings suggest high accuracy and real-time adaptability.

Bayro-Corrochano et al. [17] introduced Quaternion Spiking Neural Networks (QSNNs) and Quaternion Quantum Neural Networks (QQNNs) for control and pattern recognition tasks, thereby addressing a significant gap in quaternion-based feed-forward neural architectures. QSNNs update parameters online to guide robotic and biological systems toward target positions.

Experimental evaluations indicate strong effectiveness in controlling a robotic manipulator across diverse trajectories.

IV. QML-BASED CONTROL

The association of machine and quantum learning suits characterization and strategy-intensive issues, highlighted by diamond-based quantum technologies that enhance speed and accuracy in robotic quantum applications [18]. Petschnigg et al. explored integrating QC in robotics, identifying key applications such as sensing and perception. The authors emphasized the superior performance of quantum algorithms over classical methods and advocated further research in quantum machine learning (QML) to leverage these advantages [10]. Taghavi and Farnoosh introduced a multi-agent reinforcement learning framework that combines quantum and neuromorphic computing to enable safe, interpretable control, addressing the growing need for transparent, secure decision-making in cooperative robotic environments [19].

The ongoing advancements in Quantum Deep Reinforcement Learning highlight the potential of NISQ computing to overcome conventional robotic limitations, demonstrating improved computational efficiency and learning adaptability through quantum parallelism. This forms the foundation for integrating quantum-enhanced decision-making in autonomous robotic systems. Recent studies indicate that NISQ computing can address several classical limitations in artificial intelligence for reinforcement learning and control. Cheng et al. [20] reported single-qubit gate errors between 0.1% and 1% and two-qubit gate errors near 1%, highlighting ongoing challenges for practical quantum deployment and underscoring the need for Quantum Error Correction (QEC) to enhance robustness in quantum-assisted robotic navigation and learning. Brandhofer et al. [21] examined hardware variability across superconducting, trapped-ion, and photonic platforms, finding single-qubit fidelities of 99.9% and two-qubit fidelities of 99.4%, with superconducting qubit coherence times from 15 to 76 μ s. Their findings suggest that hybrid quantum-classical approaches, such as the Quantum Approximate Optimization Algorithm (QAOA) and the Variational Quantum Eigensolver (VQE), are essential for advancing quantum-enabled robotic perception and learning. To address hardware limits, Acampora et al. [22] proposed a Distributed NISQ framework interconnecting multiple processors, achieving a 100% success rate in distributed Grover's search versus 2% in non-distributed execution. This approach reduces circuit depth from $\sim$ 200 to 4 and boosts execution speed by 50%, benefiting real-time robotic control and optimization. Patel and Tiwari [23] introduced VERITAS, an error mitigation framework improving quantum computation reliability. Experiments on IBM Q processors show VERITAS lowering Bernstein-Vazirani algorithm errors from $\sim$ 40% to $<$ 3% and Quantum Phase Estimation errors from $\sim$ 12% to $<$ 5%, significantly enhancing quantum-assisted decision-making in robotics. Further advances in quantum gate architectures will continue to strengthen quantum-assisted robotic learning and control.

These findings apply to quantum-assisted robotic learning, where path-planning algorithms can be enhanced through stable entanglement-based control. Although NISQ-era devices remain limited by hardware noise, error rates, and coherence times,

recent advances in distributed QC, error mitigation, and robust entangling gates will enhance quantum-assisted decision-making and navigation. Bar et al. introduced a hybrid quantum method for Deep Reinforcement Learning (DRL) in navigation, reaching a 91% target success rate. Their study demonstrated QC's advantages in reducing computational constraints of DRL and provided a comparative analysis with classical solutions, indicating scalable quantum methods in robotics and setting a standard for quantum-enhanced computational tasks [24].

V. OTHER QUANTUM ALGORITHMS-BASED CONTROL AND OPTIMIZATION

Chella et al. sought speed-ups in robotic path planning by improving decision-making in complex environments. Due to limitations in quantum hardware and simulation, they emphasized practical testing on real quantum devices [25]. A novel method of robotic kinematics based on quaternion-Pauli gate equivalence was also proposed to enhance the robotic controller [26]. Pusuluri et al. used quantum-inspired firefly algorithms to coordinate swarms of autonomous delivery robots in logistics environments, significantly improving routing and decision-making under dynamic, uncertain conditions, consistent with this section's focus on decentralized optimization [27]. Recent researches prove quantum-behaved sliding-mode control (QSMC) to enhanced precision and energy efficiency in industrial robots [28, 29].

Similarly, Qian et al. developed optimal path-planning methods for two-wheeled self-balancing robots using QPSO, achieving rapid convergence and high precision. The quantum version of Particle Swarm Optimization enables broader applications and improved algorithmic efficiency in robotic path planning [30]. Fazilat and Zioui proposed a QPSO-based framework for dynamic parameter identification in a high-degree-of-freedom industrial robotic arm, extracting accurate mass and inertia parameters from CAD models optimized via QPSO. Results validated under sliding mode control showed QPSO produced lower mean absolute percentage errors (0.43% for mass centers, 0.76% for inertia tensors), smoother torque profiles, and more stable energy consumption than classical PSO. These outcomes demonstrate QPSO's effectiveness for high-dimensional nonlinear dynamics and its practicality in enhancing industrial robotic system robustness and efficiency [31]. Maity et al. applied quantum-inspired PSO (QPSO) for aerial robot path planning, surpassing traditional PSO in convergence speed and robustness, reinforcing the benefits of quantum-inspired decision-making in robotic autonomy [32].

Transitioning from individual to collective systems, Chella et al. presented a quantum algorithm for swarm robotics, demonstrating its effectiveness in search and rescue scenarios. This approach for deploying quantum-based algorithms in real robotic swarms supports broader applications of quantum methods in dynamic environments [33]. Such progress highlights these algorithms' potential to enhance individual performance and improve collective coordination in complex settings. Lathrop et al. developed a parallel quantum procedure, Parallel Quantum Rapidly-Exploring Random Trees (Pq-RRT), to improve quantum motion planning [34]. It used quantum amplitude amplification to boost search efficiency, enabling simultaneous quantum searches of a state database and

optimizing the discovery of dynamically reachable states without obstacles.

As QC advances in dynamic robotics and adaptive systems, Luo et al. [35], Xu et al. [36], and Dian et al. [37] focused on refining robotic trajectory planning using improved and hybrid quantum algorithms to address complex robotic movements, such as joint space trajectory planning for quadruped robots. They highlighted the need for adaptive optimization strategies applicable to various robotic tasks. Expanding on these developments, Fernandes et al. optimized path-planning algorithms, showing how quantum-behaved methods adapt to environmental changes and enhance trajectory planning [38].

The shift from ground-based robotic systems to specialized environments like space and underwater represents a major expansion in the scope and complexity of QC applications in robotics. Li et al. [39] applied QC to space-based systems and autonomous underwater vehicle docking, demonstrating quicker convergence, higher accuracy, and improved efficiency in challenging conditions such as space and oceans.

VI. PRACTICAL APPLICATIONS OF QUANTUM CONTROL IN ROBOTICS

In exploring practical control applications, Dong et al. [40] presented an algorithm optimizing quantum control systems under uncertainty. The algorithm achieved a fitness of 0.9798 in quantum ensemble control and 0.9988 in quantum networks, outperforming traditional methods. Mahanti et al. [41] developed a quantum circuit for Braitenberg vehicles, demonstrating deterministic behavior enhancing safety in robotic flight and gaming. Using IBM's 5-qubit chip "ibmqx4," they provided practical proof of quantum technology in real settings. Deng et al. [42] explored quantum circuits to enhance humanoid robots' entertainment value in theatrical sets, introducing a quantum automaton model enabling less predictable, more engaging behaviors. Their circuits included an Einstein Podolsky Rosen (EPR)-based quantum circuit showing ~50% probability of varied hand gestures versus the classical circuit's predictable outcomes. The study detailed servo motor control and performance challenges, highlighting the model's complexity. Nemchaninov et al. [43] combined QC and robotic prosthetics by developing an intelligent arm based on real-time Electroencephalography (EEG) analysis, a noninvasive neuroimaging method measuring brain activity to adapt to emotional states, advancing personalized, emotionally responsive aids. Mamaeva et al. [44] integrated EEG data for real-time navigation, enabling autonomous robots to avoid obstacles while classifying emotional states, demonstrating the precision of quantum-enhanced cognitive control in adaptive robotics. Reshetnikova et al. [45] validated a quantum fuzzy inference-based Intelligent Control System (ICS) for distributed robotic platforms, showing robustness under unpredictable conditions in a "cart-pole" system. Mei et al. [46] demonstrated QC's potential in robotic dance creation via quantum simulations with GANs and digital twins, improving movement generation efficiency and accuracy. Chen et al. [47] integrated QPSO with Model Predictive Control (MPC), reducing tracking errors by 43% in Cable-Driven Continuum Robots (CDCRs), confirming its effectiveness in sensitive environments.

In adaptive controllers, Hanna et al. [16] employed a quantum neural network to reduce tunable parameters, enhancing stability in a two-wheel mobile robot. Similarly, Singh and Sloth [48] developing a Type 2 Quantum Fuzzy Neural Network (ET2QFNN) for motion control, achieving an RMSE of 0.1618, while the Proportional-Derivative Controller (PD) controller averaged 0.2631 and the Improved Sliding Mode Control (iSMC) 0.3278, clearly outperforming traditional controllers. Wan et al. [49] optimized fractional-order PID controllers with a quantum genetic algorithm, reducing overshoot and settling times during sea trials. Yang [50] applied quantum genetic algorithms to soccer robot path planning, obtaining a 4.0253-meter path versus 4.3813 meters using classical methods. Varma et al. [51] addressed security in mobile agents with post-quantum cryptography, observing minimal trade-offs even on Raspberry Pi Zero W. Yin et al. [52] enhanced robotic machining via an improved quantum swarm optimization for optimized Support Vector Machine (SVM) parameters, better managing nonlinearities in milling forces. Reshetnikova et al. [45] applied quantum fuzzy inference for robust control under uncertainty, while Khoshnoud and Ghazinejad [53] investigated quantum communication alignment for efficient network management. Despite computational and integration challenges, these studies present innovative frameworks enhancing precision, communication, and control, advancing scalable, practical robust control applications.

VII. DISCUSSION

The comparative analysis of quantum-based control strategies reveals that each method, Quantum, QNN, QPSO, and QML, offers unique advantages and trade-offs in terms of accuracy, efficiency, and scalability. QNN-based controllers exhibit exceptional modeling precision and adaptability, typically achieving RMSEs that are 8–12% lower and validation errors that are approximately 84% lower than those of classical neural networks, as detailed in references [15] and [16]. However, these enhancements come with increased computational complexity and longer convergence times due to the iterative nature of quantum weight training. In contrast, QPSO frameworks demonstrate faster convergence and smoother torque behavior in industrial manipulators [31], surpassing traditional Particle Swarm Optimization (PSO) in mean absolute percentage errors (0.43% for mass centers and 0.76% for inertia tensors). Nevertheless, their scalability is limited when applied to high-dimensional robotic systems requiring continuous adaptation. On the other hand, QML-based hybrid algorithms show notable learning flexibility and scalability, particularly in multi-agent and cognitive robotic environments [19,24]. However, their effectiveness is inhibited by challenges such as hardware noise, coherence loss, and the limited quantity of qubits in current NISQ devices [20-22]. Table I provides a concise comparative summary of the main quantum-based control techniques, highlighting their key strengths and limitations.

Quantitatively, most reported improvements in tracking accuracy (ranging from 5% to 15%), torque smoothness (up to 40%), and convergence speed (between 20% and 50%) have been observed under simulation or hybrid quantum-classical conditions rather than on actual quantum hardware. Translating

these simulation-based results into real-time robotic control remains a challenge due to computational overhead, circuit depth, and qubit allocation requirements. Moreover, the latency associated with the exchange of quantum and classical data, coupled with the overheads of quantum error correction, can diminish the theoretical quantum speedups by approximately 10% to 20% in practical implementations. These findings underscore the need for lightweight, noise-tolerant, and hardware-aware quantum control algorithms that effectively balance precision with real-time feasibility. Despite promising simulation results, the scarcity of experimental implementations on real quantum hardware continues to constrain progress toward practical and scalable robotic systems. Fortunately, hybrid quantum-classical frameworks, with their practicality and potential, emerge as the most reassuring near-term solution for scalable and fault-tolerant robotic control within the constraints imposed by current NISQ-era quantum technologies.

TABLE I. COMPARATIVE SUMMARY OF QUANTUM-BASED CONTROL TECHNIQUES FOR ROBOTICS

Technique	Key Strengths	Main Limitations
QNN	8–12 % lower RMSE, ≈ 84 % lower validation error [15],[16]; accurate and robust for nonlinear control.	High training cost; slow convergence; limited real-time feasibility under current hardware.
QPSO	20–40 % faster convergence, < 1 % parameter error [31],[32]; smoother torque and efficient energy use.	Poor scalability for high-DOF robots; needs frequent retuning; mainly offline-feasible.
QML / Hybrid DRL	≈ 91 % success, ≈ 50 % faster learning [19],[24]; strong scalability in multi-agent tasks.	Hardware noise (0.1–1 %); coherence limits; 10–20 % performance loss from error correction.
QSMC	Improved precision & energy efficiency in industrial arms [28],[29]; fault-tolerant control.	Limited quantum-hardware interfaces; hybrid loops required; untested on physical robots.
Quantum Genetic / Fuzzy	Lower overshoot & settling time [49],[50]; smoother motion & better fault tolerance [48].	Sensitive to noise; moderate real-time feasibility; low-speed use only.
Classical Control / AI	Fully scalable, reliable, low-latency control on classical hardware.	Less precise in nonlinear / stochastic systems; lacks adaptability & quantum speedups.

VIII. CONCLUSION

This review summarizes advances in integrating QC with robotic control, emphasizing how quantum-inspired algorithms, QNN, QML, and QPSO, enhance precision, robustness, and energy efficiency. Quantum-based methods outperform traditional strategies in addressing nonlinearities and high-dimensional uncertainties. Yet most results remain limited to simulations with minimal validation on real hardware, hindering the translation of theory into practical, scalable implementations. Future research should bridge this gap through hybrid quantum–classical architectures, fault-tolerant quantum controllers, and distributed quantum frameworks capable of

real-time robotic control. Addressing these challenges will enable the transition from simulation to hardware-level control, realizing intelligent, adaptive, and energy-efficient robotics powered by quantum technologies.

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